



EarthBank by AuScope: Building FAIR research data infrastructure for the global geochemical community

Angus Nixon¹, Bryant Ware², Brent McInnes², Fabian Kohlmann³, Moritz Theile³, Wayne Noble³, Yoann Gréau⁴, Hayden Dalton⁵, Halimulati Ananuer⁴, Malcolm McMillan⁵, and Ashley Savelkouls⁵

¹Department of Earth Science, The University of Adelaide, Adelaide, Australia

²John de Laeter Centre, Curtin University, Bentley, Australia

³Lithodat Pty Ltd, Melbourne, Australia

⁴Research School of Earth Sciences, The Australian National University, Canberra, Australia

⁵School of Geography, Earth and Atmospheric Sciences, The University of Melbourne, Melbourne, Australia

The geochemical community increasingly generates and requires large volumes of analytical data from a wide array of acquisition methods, analytical scales, and sample types in order to address broad research applications. Resulting datasets are commonly collected and reported through non-standardised protocols and reporting formats, if indeed standards are applied at all, which inhibits easy sharing of data during collaborative research projects or repurposing of legacy data. Existing repository services do not presently satisfy requirements for Findable, Accessible, Interoperable and Reusable (FAIR) data, and especially contain significant flaws as to the reuse and interoperability of geochemical data. Generalist repositories such as Zenodo or Figshare do not provide consistent data structures or curation, hence data held within these services is highly variable with regard to format, parameters reported and potentially quality. While domain repositories commonly do implement internally consistent data formats and a level of curation, data within repositories is gathered from published sources which may be incomplete or unstructured, and hence often lack the complete information (metadata) required to appropriately describe the data and allow it to be confidently reused.

To truly unlock the potential of the ever expanding wealth of geochemical data and meet FAIR requirements, improvements to the data infrastructure landscape are clearly required. The AuScope Geochemistry Network (AGN) is an Australian-based collaboration of geoscientists producing bespoke data resources and infrastructure for the international community to capture, normalise, and share geochemical data resources. These resources include best practice data reporting schema and vocabularies for a variety of data types, produced through collaborations with expert advisory groups and, where available, following or expanding on existing international community recommendations. These data resources have been implemented to the EarthBank platform (formerly AusGeochem), an open web service designed by the AGN to capture, share, store and evaluate geochemical data and metadata. Unlike many other services, researchers are able to upload data prior to publication which can assist both in allowing researchers to compare their data with other existing resources prior to submission, but importantly also improves the

likelihood of capturing the full data and metadata associated with analyses required for reuse. Once data is uploaded to this service it may be associated with a dataset DOI to support data access requirements for publication, in order to streamline the publication process and provide a domain specific repository for supplemental data. Data models for U/Pb, fission track, (U-Th-Sm)/He, $^{40}\text{Ar}/^{39}\text{Ar}$ and inorganic major and trace geochemistry data types are presently implemented within EarthBank, allowing users to freely upload generated research data for these systems, or explore and integrate existing datasets. Best practice templates for upload are openly available through the EarthBank platform, and vocabularies are openly discoverable through the Research Vocabularies Australia (RVA) service. These resources may be used not only to upload data, but also to develop cross-walks for machine-to-machine interoperability with other repository services to build a global FAIR compliant infrastructure required to maximise data access and improve research outcomes.